

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Circles

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03

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C**

C. $x - 4^2 + y - 9^2 = 16$

Choice C is correct. The graph of the equation $x - h^2 + y - k^2 = r^2$ in the xy -plane is a circle with center h, k and a radius of length r . The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the xy -plane intersects the y -axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the y -axis must be equal to the length of the circle's radius. It follows that the x -coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $x - h^2 + y - k^2 = r^2$ is a circle that intersects the y -axis at exactly one point, then $r = h$ must be true. The equation in choice C is $x - 4^2 + y - 9^2 = 16$, or $x - 4^2 + y - 9^2 = 4^2$. This equation is in the form $x - h^2 + y - k^2 = r^2$, where $h = 4$, $k = 9$, and $r = 4$, and represents a circle in the xy -plane with center 4, 9 and radius of length 4. Substituting 4 for r and 4 for h in the equation $r = h$ yields $4 = 4$, or $4 = 4$, which is true. Therefore, the equation in choice C represents a circle in the xy -plane that intersects the y -axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the y -axis at any point.

Choice B is incorrect. This is an equation of a circle that intersects the x -axis, not the y -axis, at exactly one point.

Choice D is incorrect. This is the equation of a circle with the center located on the y -axis and thus intersects the y -axis at exactly two points, not exactly one point.

Q2

A

A. 5

Choice A is correct. One way to find the radius of the circle is to rewrite the given equation in standard form, $(x-h)^2 + (y-k)^2 = r^2$, where (h,k) is the center of the circle and the radius of the circle is r . To do this, divide the original equation, $2x^2 - 6x + 2y^2 + 2y = 45$, by 2 to make the leading coefficients of x^2 and y^2 each equal to 1: $x^2 - 3x + y^2 + y = 22.5$. Then complete the square to put the equation in standard form. To do so, first rewrite $x^2 - 3x + y^2 + y = 22.5$ as $(x^2 - 3x + 2.25) - 2.25 + (y^2 + y + 0.25) - 0.25 = 22.5$. Second, add 2.25 and 0.25 to both sides of the equation: $(x^2 - 3x + 2.25) + (y^2 + y + 0.25) = 25$. Since $x^2 - 3x + 2.25 = (x - 1.5)^2$, $y^2 + y + 0.25 = (y + 0.5)^2$, and $25 = 5^2$, it follows that $(x - 1.5)^2 + (y + 0.5)^2 = 5^2$. Therefore, the radius of the circle is 5.

Choices B, C, and D are incorrect and may be the result of errors in manipulating the equation or of a misconception about the standard form of the equation of a circle in the xy -plane.

Q3**A**

A. $x + 2^2 + y + 6^2 = 49$

Choice A is correct. According to the graph, the center of circle A has coordinates $-2, 0$, and the radius of circle A is 3. It's given that circle B is the result of shifting circle A down 6 units and increasing the radius so that the radius of circle B is 2 times the radius of circle A. It follows that the center of circle B is 6 units below the center of circle A. The point that's 6 units below $-2, 0$ has the same x -coordinate as $-2, 0$ and has a y -coordinate that is 6 less than the y -coordinate of $-2, 0$. Therefore, the coordinates of the center of circle B are $-2, 0 - 6$, or $-2, -6$. Since the radius of circle B is 2 times the radius of circle A, the radius of circle B is 23. A circle in the xy -plane can be defined by an equation of the form $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the radius of the circle is r . Substituting -2 for h , -6 for k , and 23 for r in this equation yields $x - 2^2 + y - 6^2 = 23^2$, which is equivalent to $x + 2^2 + y + 6^2 = 2^2 3^2$, or $x + 2^2 + y + 6^2 = 49$. Therefore, the equation $x + 2^2 + y + 6^2 = 49$ defines circle B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation defines a circle that's the result of shifting circle A up, not down, by 6 units and increasing the radius.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**11**

The correct answer is 11. A circle with equation $(x - a)^2 + (y - b)^2 = r^2$, where a , b , and r are constants, has center (a, b) and radius r . Therefore, the radius of the given circle is $\sqrt{121}$, or 11.

Q5

D

D. $(3, 2)$

Choice D is correct. The circle with equation $(x+3)^2 + (y-1)^2 = 25$ has center $(-3, 1)$ and radius 5. For a point to be inside of the circle, the distance from that point to the center must be less than the radius, 5. The distance between $(3, 2)$ and $(-3, 1)$ is

$$\sqrt{(-3-3)^2 + (1-2)^2} = \sqrt{(-6)^2 + (-1)^2} = \sqrt{37}, \text{ which is greater than 5. Therefore, } (3, 2)$$

does NOT lie in the interior of the circle.

Choice A is incorrect. The distance between $(-7, 3)$ and $(-3, 1)$ is

$$\sqrt{(-7+3)^2 + (3-1)^2} = \sqrt{(-4)^2 + (2)^2} = \sqrt{20}, \text{ which is less than 5, and therefore } (-7, 3) \text{ lies}$$

in the interior of the circle. Choice B is incorrect because it is the center of the circle. Choice C is incorrect because the distance between $(0, 0)$ and $(-3, 1)$ is

$$\sqrt{(0+3)^2 + (0-1)^2} = \sqrt{(3)^2 + (1)^2} = \sqrt{8}, \text{ which is less than 5, and therefore } (0, 0) \text{ in the}$$

interior of the circle.

Q6**B****B.** 8

Choice B is correct. The standard form of an equation of a circle in the xy -plane is $x - h^2 + y - k^2 = r^2$, where the coordinates of the center of the circle are h, k and the length of the radius of the circle is r . For the circle in the xy -plane with equation $x - 5^2 + y - 3^2 = 16$, it follows that $r^2 = 16$. Taking the square root of both sides of this equation yields $r = 4$ or $r = -4$. Because r represents the length of the radius of the circle and this length must be positive, $r = 4$. Therefore, the radius of the circle is 4. The diameter of a circle is twice the length of the radius of the circle. Thus, 24 yields 8. Therefore, the diameter of the circle is 8.

Choice A is incorrect. This is the radius of the circle.

Choice C is incorrect. This is the square of the radius of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**16**

The correct answer is 16. An equation of a circle in the xy -plane can be written as $x - t^2 + y - u^2 = r^2$, where the center of the circle is t, u , the radius of the circle is r , and where t, u , and r are constants. It's given that the equation of circle A is $x + 5^2 + y - 5^2 = 4$, which is equivalent to $x + 5^2 + y - 5^2 = 2^2$. Therefore, the center of circle A is $-5, 5$ and the radius of circle A is 2. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $-5, 5$ and the radius of circle B is 22, or 4. Substituting -5 for t , 5 for u , and 4 for r into the equation $x - t^2 + y - u^2 = r^2$ yields $x + 5^2 + y - 5^2 = 4^2$, which is equivalent to $x + 5^2 + y - 5^2 = 16$. It follows that the equation of circle B in the xy -plane is $x + 5^2 + y - 5^2 = 16$. Therefore, the value of k is 16.

Q8**B**

B. $(x - 19)^2 + (y - 18)^2 = 81k^2$

Choice B is correct. The equation of a circle in the xy -plane can be written as

$(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that this circle has a center at $(19, 18)$ and a radius of $9k$. Substituting 19 for h , 18 for k , and $9k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 19)^2 + (y - 18)^2 = (9k)^2$, or $(x - 19)^2 + (y - 18)^2 = 81k^2$. Therefore, the equation that represents this circle is $(x - 19)^2 + (y - 18)^2 = 81k^2$.

Choice A is incorrect. This equation represents a circle with radius $9\sqrt{k}$, not $9k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{9} k$, not $9k$.

Q9**B**

B. 13π

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\overline{ADC}$ is subtended by a central angle with measure 100° . Since the sum of the measures of the angles about a point is 360° , it follows that arc $\overline{ABC}$ is subtended by a central angle with measure

$360^\circ - 100^\circ = 260^\circ$. If s is the length of arc $\overline{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi} = \frac{260}{100}$.

Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi} = \frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi} = \frac{13}{5}$ by 5π yields $s = 13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\overline{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\overline{ABC}$, not its full length.

Q10

D

D. 24

Choice D is correct. It's given that the measure of arc AB is 45° and the length of arc AB is 3 inches. The arc measure of the full circle is 360° . If x represents the circumference, in inches, of the circle, it follows that $\frac{45^\circ}{360^\circ} = \frac{3 \text{ inches}}{x \text{ inches}}$. This equation is equivalent to $\frac{45}{360} = \frac{3}{x}$, or $\frac{1}{8} = \frac{3}{x}$. Multiplying both sides of this equation by $8x$ yields $1(x) = 3(8)$, or $x = 24$. Therefore, the circumference of the circle is 24 inches.

Choice A is incorrect. This is the length of arc AB .

Choice B is incorrect and may result from multiplying the length of arc AB by 2.

Choice C is incorrect and may result from squaring the length of arc AB .

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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